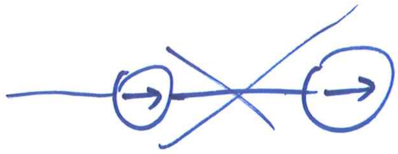
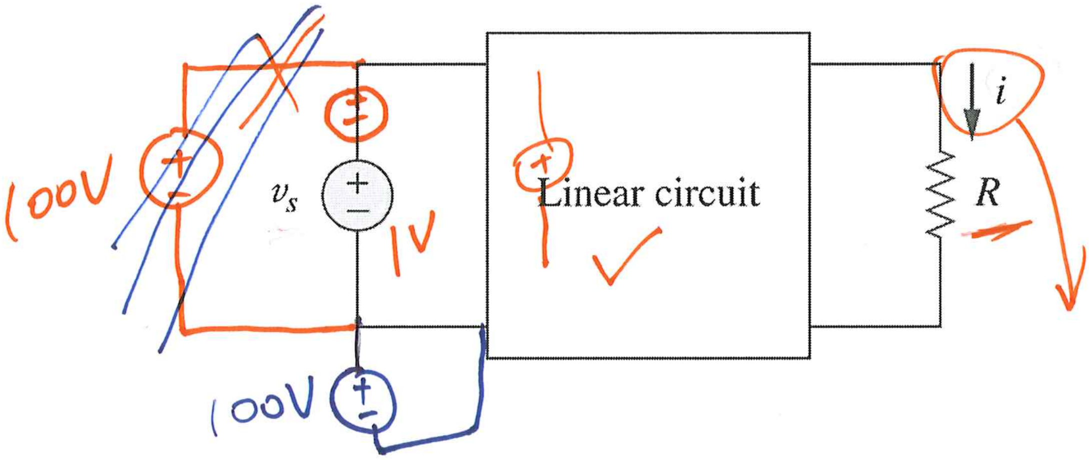


L#13: Linearity and superposition



If $v_s = 10V \implies i = 2$ A

then:

$v_s = 1V \implies i =$ ___0.2___ A

$v_s = 100V \implies i =$ ___20___ A

try add above two

20.2

Two properties

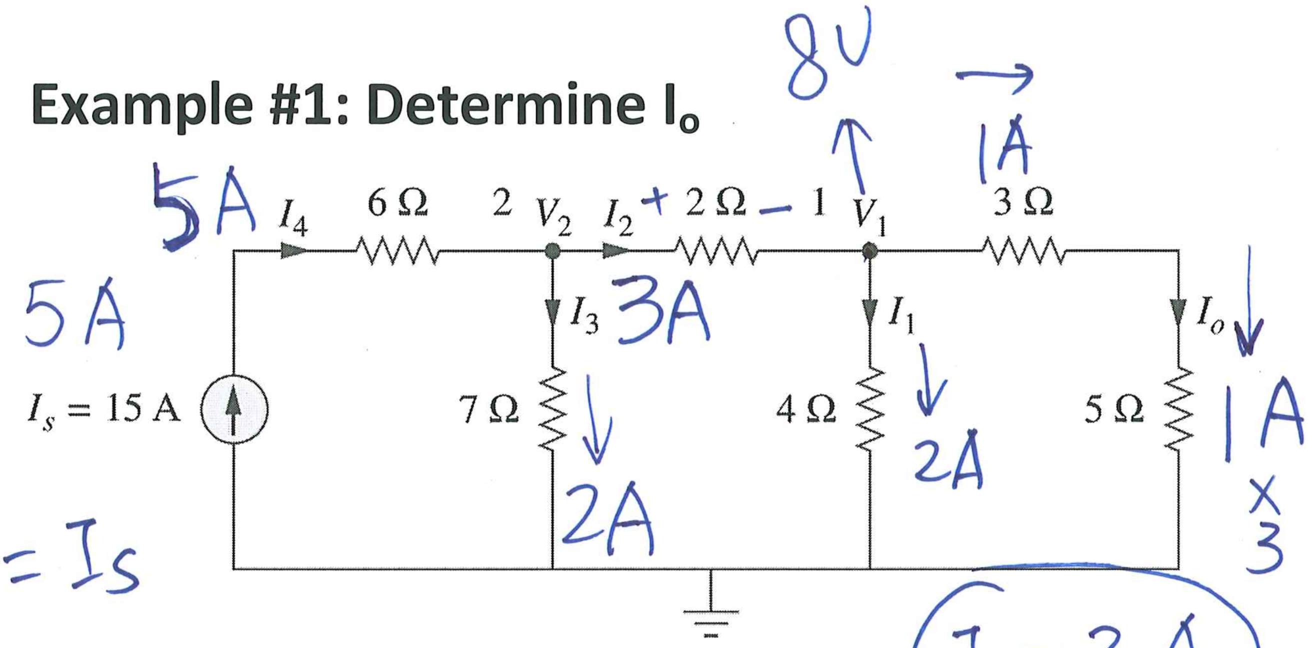
- Homogeneity (scaling)

v $V = iR$; $kV = ki\,R$

- Additivity

v_1 and v_2 $\left.\begin{array}{l} V_1 = i_1 R. \\ V_2 = i_2 R \end{array}\right\} \Rightarrow V_1 + V_2 = (i_1 + i_2)R$

Example #1: Determine I_o



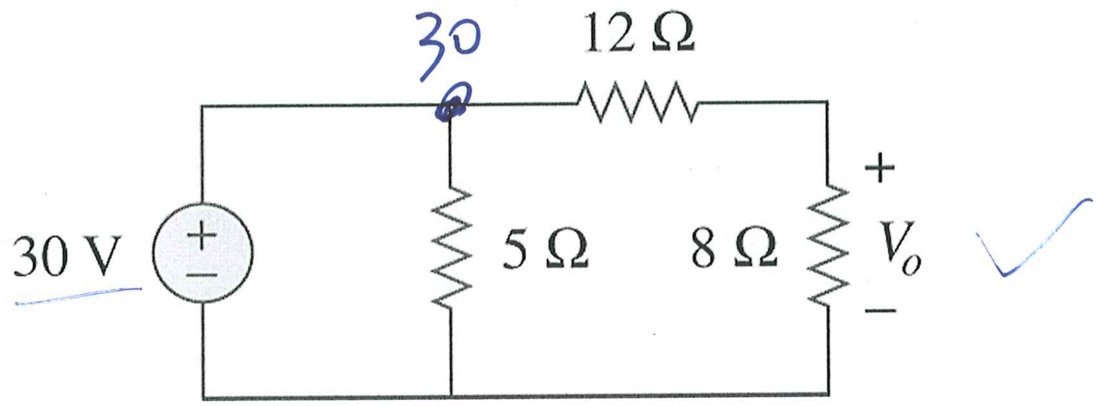
$I_4 = I_s$

$I_o = 3A$ actual

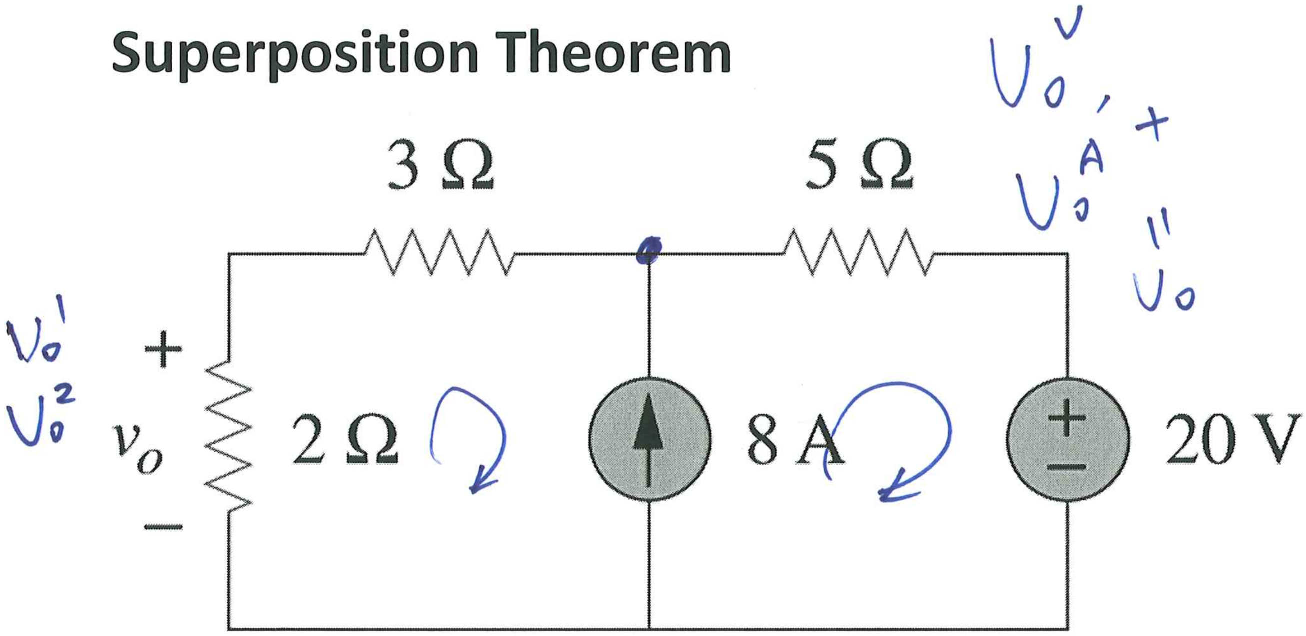
Assume $I_o = 1A$

$V_1 = 8V, \quad I_1 = 2A, V_2 = 14V$

Example #2: Determine V_o.

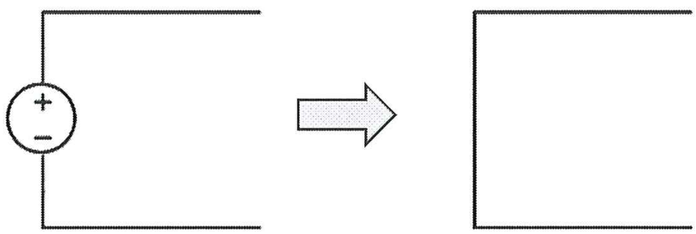
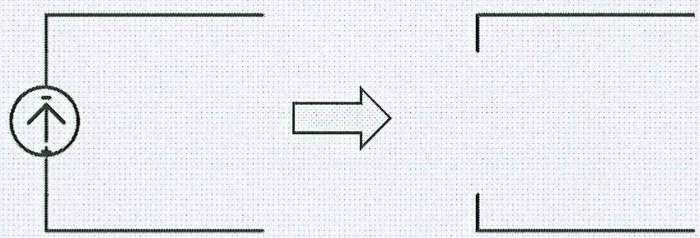


Superposition Theorem



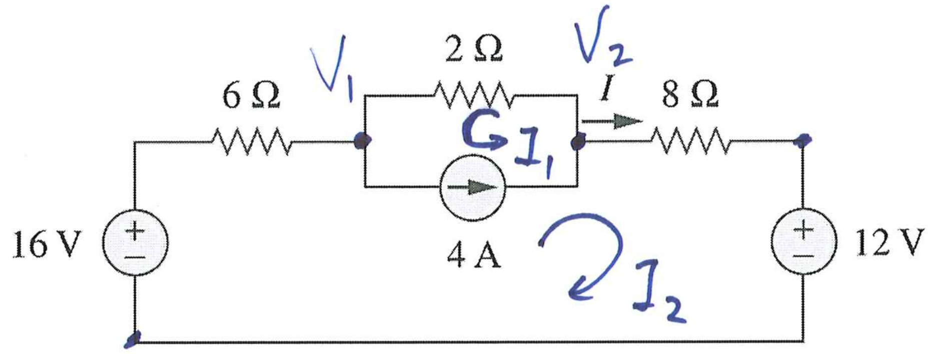
1. Turn off; 2. Repeat; 3. Sum.

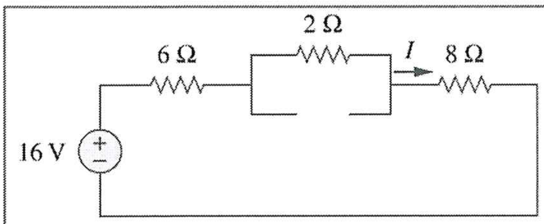
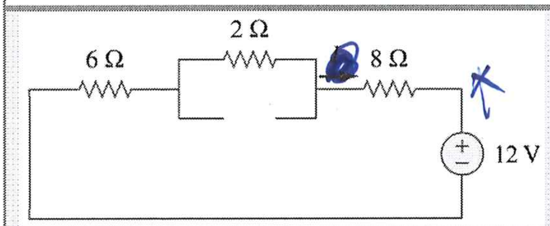
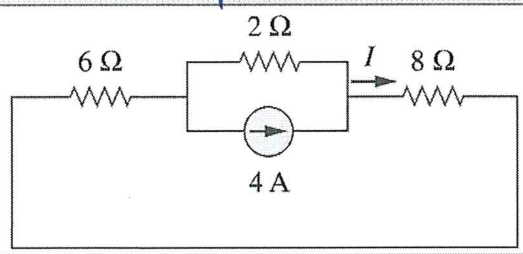
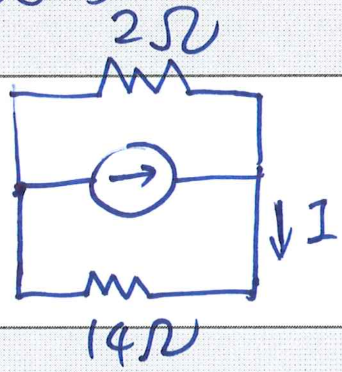
1. Turn off **INDEPENDENT** sources

Voltage source → Short circuit	
Current source → Open circuit	
Dependent Source	Leave as it is

2. Repeat Step 1 with analysis
3. Summation Addility

Example #1: Find I.



 $I_1 = \frac{16}{16} = 1A$	$I_1 = \frac{16}{16} = 1A$
 $I_2 = -\frac{12}{16} = -0.75A$	$I_3 = 4A \cdot \frac{2}{16 = 2+14}$ $= 0.5A$
	
Total $I = I_1 + I_2 + I_3 = 1 - 0.75 + 0.5 = 0.75A$	

$$M1: \quad I_1 + I_2 = 4$$

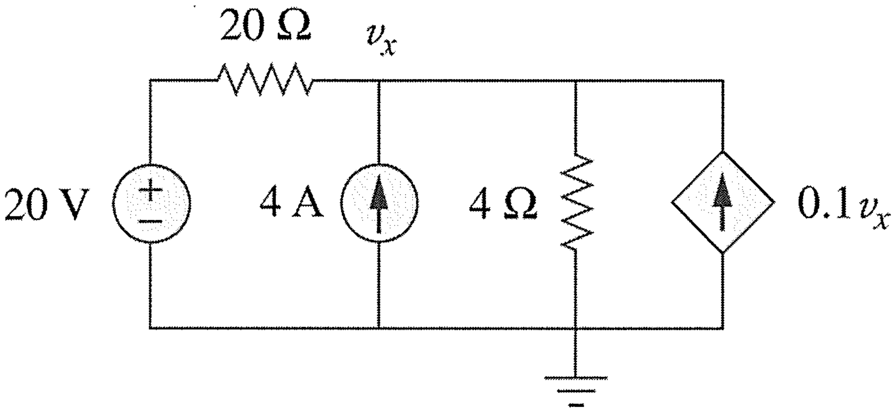
$$M2: \quad -16 + 6I_2 - 2I_1 + 8I_2 + 12 = 0$$

$$\Rightarrow I = 4 - I_1 = 0.75A$$

$$V_1: \quad \frac{V_1 - 16}{6} + 4 + \frac{V_1 - V_2}{2} = 0$$

$$V_2: \quad I = \frac{V_2 - 12}{8} = 4 + \frac{V_1 - V_2}{2} = 0.75A$$

Example #2: Find v_x.



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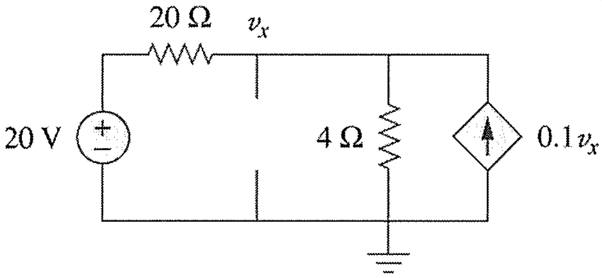
<tr><td>
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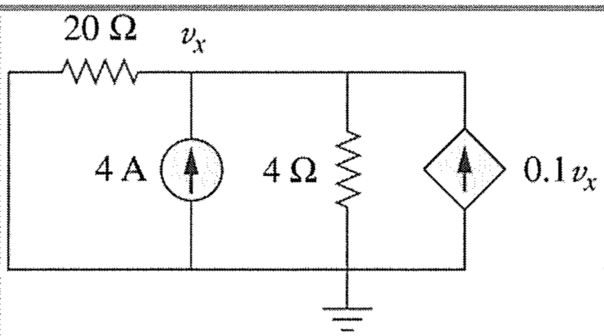
<tr><td>
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